

liquidity, macroeconomic, and aggregate financial condition risks account for up to 40% of all credit spread changes. Longstaff, Mithal, and Neis (2005) show that almost all of the nondefault components of credit spreads are attributable to liquidity. Liquidity certainly played a very important role in the returns of corporate bonds during the financial crisis. Figure 9.17 plots the Baa-Treasury yield spread (left-hand axis) together with the on-the-run/off-the-run liquidity factor (right-hand axis) from January 2006 to October 2009. (Figure 9.17 for corporate spreads is similar to Figure 9.12 for Treasury term spreads.) Movements in the credit spread uncannily mirror the changes in illiquidity; the correlation between the two series is 96%!²⁹ *Corporate bonds have large exposures to illiquidity risk.*

5. U.S. Downgrade Redux

In August 2011 when the United States was downgraded for the first time in history by Standard & Poor's, Treasury yields actually fell. If U.S. credit risk had increased with everything else equal, investors should have demanded a higher risk premium for holding Treasuries, resulting in higher yields rather than lower ones. An important factor for bond prices, especially Treasuries, is illiquidity and volatility risk. U.S. Treasuries have a flight-to-quality characteristic that investors embrace during bad times. In the turmoil of the ongoing European debt crisis, investors poured money into Treasuries as a safe haven, pushing down Treasury yields despite the U.S. downgrade.

Liquidity, however, turns out to be a secondary characteristic of Treasury bond prices. The most important factor for all fixed income instruments is the level factor—which simply acts to shift all yields up or down. The level factor is well described by the actions of the Fed reacting to economic growth and inflation changes. As economic growth accelerates or inflation rises, the Fed raises interest rates to cool the economy. Thus, bond prices reflect economic factor risk, especially inflation risk, along with the risk of unanticipated monetary policy shocks.

In this chapter, we treated Treasuries as risk free. As the U.S. downgrade in August 2011 implied, U.S. Treasuries may not be totally risk free. We deal with risky sovereign debt in chapter 14, including the implications for optimal portfolio choice.

Corporate bonds are exposed to the same risks as risk-free debt but also reflect default risk. The credit risk premium, however, has been fairly modest and much smaller than the equity risk premium. Investing in corporate bonds turns out to be equivalent to a short volatility strategy; as volatility increases, credit spreads also increase. Thus, credit strategies are highly exposed to volatility risk.

²⁹ Friewald, Jankowitsch, and Subrahmanyam (2012) show that the impact of liquidity on corporate bond spreads increases during periods of crises, especially for junk bonds.